ATUL KUMAR TRIPATHI

Mathematics Researcher • Knowledge Representation • Quantitative Modelling

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Academic Website • LinkedIn • GitHub • Google Scholar • ORCID

RESEARCH INTERESTS

Knowledge Representation and Knowledge Graphs; Machine Learning and Data Mining; Natural Language Processing; Statistical Learning and Inference; Time-Series Analysis and Financial Econometrics; Mathematical Modelling.

EDUCATION

Indian Institute of Technology Delhi (IIT Delhi) 2023–2025
M.Sc. Mathematics **CGPA: 8.038/10**

University of Allahabad 2020–2023
B.Sc. Mathematics, Physics and Computer Science **79.03%**

RESEARCH EXPERIENCE

Cryptocurrency Markets and Geopolitical Risk: A Study of the Israel–Gaza Conflict 2025–Present
Replication and extension of a high-frequency event-study framework

- Replicated and extended a high-frequency event-study framework to examine how geopolitical shocks affected **cryptocurrency and traditional financial markets** during the Israel–Gaza conflict.
- Constructed an **Hourly War Risk Index (HWRI)** from time-stamped geopolitical news to distinguish escalation and de-escalation shocks.
- Matched geopolitical events with hourly market data, consolidating **3,395 reports into 2,835 unique events**; estimated **AR, CAR and CAAR** over 1–48 hour windows.
- Applied **Welch's t-tests, effect-size analysis and Benjamini–Hochberg FDR correction**; results showed **heterogeneous, short-lived asset-specific responses** with no persistent market-wide relationship over the full study period.

Knowledge Representation for Healthcare Systems: A Comparative Analysis of Performance of Different Representation Paradigms 2024–2026
Published research; Springer conference proceedings

- Developed a unified framework comparing **seven knowledge-representation paradigms** for healthcare question answering across increasing levels of reasoning complexity.
- Constructed a benchmark of **47 cardiovascular patient records** and five clinical queries spanning Boolean, relational, multi-relational and contextual reasoning.
- Evaluated the paradigms using **query coverage, reasoning depth and explainability**; Knowledge Graphs achieved **complete coverage** of all benchmark queries while supporting multi-hop and contextual reasoning.

PUBLICATION

Atul Kumar Tripathi, Puja Minodji Thakre, and Niladri Chatterjee.
Knowledge Representation for Healthcare Systems: A Comparative Analysis of Performance of Different Representation Paradigms.
Intelligent Computing: Proceedings of the 2026 Computing Conference, Volume 2, Lecture Notes in Networks and Information Systems, vol. 1950, pp. 38–54, Springer, 2026.
DOI: 10.1007/978-3-032-24807-7_4

ACADEMIC DISTINCTIONS

CSIR–UGC NET, Mathematical Sciences — Qualified for PhD, Lectureship and Assistant Professorship.

SCHOLARSHIPS & ACADEMIC LEADERSHIP

- **INSPIRE Scholarship & Fellowship** — Awarded during B.Sc. and extended through M.Sc. at IIT Delhi, including summer research support.
- **Merit-cum-Means Scholarship, IIT Delhi** — Awarded on academic merit with full tuition-fee waiver and monthly financial support.
- **Department Convenor, Mathematics, IIT Delhi (2024–2025)** — Elected student representative for the M.Sc. Mathematics cohort.
- **PG Coordinator, Mathematics Society (MathSoc), IIT Delhi (2024–2025)** — Coordinated academic and sports activities

and student engagement.

- **State Academic Honour** — Honoured by the Hon'ble Chief Minister of Uttar Pradesh for securing **District Rank 7 with 91%** in the Class X Board examinations.

TECHNICAL SKILLS

Programming & Data Analysis: C, C++, Python, SQL, LaTeX;
Machine Learning & Data Mining: Regression, Classification, Data Mining
Knowledge Representation: Knowledge Graphs, Ontologies
Statistics & Quantitative Methods: Probability and Statistics, Time-Series Analysis

SELECTED COURSEWORK

M.Sc. Mathematics, IIT Delhi:
Abstract Algebra; Linear Algebra; Real Analysis; Complex Analysis; Measure and Integration; Ordinary Differential Equations; Partial Differential Equations; Functional Analysis; Topology; Numerical Analysis; Probability and Statistics; Mathematical Programming; Multiple Decision Procedures in Ranking and Selection; Computer Programming; Mathematics Behind Machine Learning; Data Mining.

PROFESSIONAL EXPERIENCE

Nine Education, Hyderabad May 2026–Present
Mathematics Faculty

- Teaching Mathematics for JEE Main and JEE Advanced cohorts; developing advanced problem-solving and assessment material.

Bhautiki Plus Ed-Tech Pvt. Ltd., Hyderabad May 2025–May 2026
Mathematics Faculty

- Taught Mathematics for JEE Main, JEE Advanced and Foundation cohorts; developed problem sets, quizzes, assessments and instructional material.

REFERENCES

- **Dr. Niladri Chatterjee** – Ex. Professor, Department of Mathematics, IIT Delhi. Co-author. Email: niladri@maths.iitd.ac.in
- **Neeraj Joshi** – Assistant Professor, Department of Mathematics, IIT Delhi. Email: njoshi@maths.iitd.ac.in